Strand 1: Number

Sub-strand 2: Number Operations

Lesson 2 (Day 4): Number Properties and Operational Laws

1. Curriculum Standard and Overview

• Strand: 1 — Number
• Sub-strand: 2 — Number Operations
• Content Standard: B9.1.2.1 — Demonstrate understanding of number properties and operational laws, and apply them to solve problems.
• Indicators:
• B9.1.2.1.2 — Determine the commutative properties of addition and multiplication.
• B9.1.2.1.3 — Use the associative property of addition and multiplication.
• B9.1.2.1.4 — Use the distributive property in solving problems.
• Scope of this lesson:
1. The commutative property — swapping numbers.
2. The associative property — re-grouping numbers.
3. Operations that break the laws (subtraction and division).
4. The distributive property — sharing a multiplier out.
5. Using the three laws for fast mental arithmetic.
6. A bridge into algebra: why these laws make expressions like 5(3-y) expand the way they do.
• Mock-paper links (this lesson feeds):
• Mock 2026 P1 Q18 — expanding 5 - 5(3-y) requires the distributive property (full inequality treatment comes in the Equations & Inequalities lesson).
• Mock 2026 P1 Q40 — dividing 12 cakes into equal packets builds on the structure of multiplication (included as an exercise).
• Mock 2026 P2 Q4(d) and Q6(a) — expanding brackets such as 2[(x+2)+(2x+1)] and (6x+2)(5x+8) is powered by the distributive property.
• Exercise quota (user override, October 2026): 5 exercise sections × 5 questions = 25 questions, no multiple choice, every question with a full worked answer. (This replaces the tracker's default 8 × 5 quota for lessons delivered in this round. Spiral review: 20% — questions 4(d)5, 5(e)4, 5(e)5 revisit earlier lessons.)

2. Preamble — What Is a "Property" or a "Law"?

Every game has rules that never change. Football: the ball must cross the line to be a goal — every match, every pitch, every referee. Mathematics has rules like that too. We call them properties or laws of operation.

A property is a fact about how numbers behave that is true always, not just sometimes. If a statement is true for every possible pair or group of numbers, it deserves the title "property". If it fails even once, it is not a property — and part of this lesson is spotting the failures.

Why should you care? Two reasons:

1. Speed. The laws are the engine of mental arithmetic. 8 × 43 × 125 looks slow, but re-group it (associative property) and it takes two seconds.
2. Algebra. Everything you will do later with brackets — expand, factorise, solve equations — is just these three laws wearing algebraic clothes. Learn them now with plain numbers, and algebra will feel familiar.

Unpacking the three key words (each is stolen from ordinary English):

Word	Everyday meaning	In mathematics
Commutative	from commute — to swap/interchange places (like commuting between Kumasi and Accra: two direction options)	You can swap the two numbers — the order does not matter
Associative	from associate — to group with/keep company	You can re-group the numbers (move the brackets) — the grouping does not matter
Distributive	from distribute — to hand out, share out (like a teacher distributing books)	You can hand out the multiplication to each number inside the bracket

3. The Commutative Property — Swap the Numbers

3.1 Addition is commutative

For any numbers a and b:

$$a + b = b + a$$

Check with real values: 7 + 5 = 12 and 5 + 7 = 12. Same sum. Put 5 000 cakes on one side and 7 on the other — swapping which one you count first does not change how many you have in total.

3.2 Multiplication is commutative

For any numbers a and b:

$$a \times b = b \times a$$

4 × 927 = 927 × 4 = 3 708. Think of an array: 4 rows of 927 dots vs 927 rows of 4 dots — different orientations, same number of dots.

3.3 Subtraction and division are NOT commutative

This is where students lose marks: assuming "it works" for subtraction and division.

$$9 - 4 = 5, \qquad 4 - 9 = -5. \qquad 5 \neq -5.$$
$$12 \div 3 = 4, \qquad 3 \div 12 = 0.25. \qquad 4 \neq 0.25.$$

Rule to hold on to: the commutative property belongs to addition and multiplication only. If you ever see a - b = b - a or a ÷ b = b ÷ a claimed as a general rule, it is false — one counterexample settles it.

4. The Associative Property — Move the Brackets

4.1 Addition is associative

For any numbers a, b, c:

$$(a + b) + c = a + (b + c)$$

The brackets only say "do this pair first". With addition you can compute any pair first and land on the same total:

$$(2 + 19) + 81 = 21 + 81 = 102, \qquad 2 + (19 + 81) = 2 + 100 = 102.$$

Notice how the second version was easier: 19 + 81 gives a round 100. This is the associative property at work as a mental maths tool — hunt for pairs that make multiples of 10.

4.2 Multiplication is associative

For any numbers a, b, c:

$$(a \times b) \times c = a \times (b \times c)$$

(8 × 5) × 20 = 40 × 20 = 800 and 8 × (5 × 20) = 8 × 100 = 800. The first grouping gave 40; the second gave a friendlier 100. Choose the grouping that produces round numbers!

4.3 The identities (special neutral numbers)

Two small properties travel with the associative property:

- Additive identity: a + 0 = 0 + a = a. Zero changes nothing.
- Multiplicative identity: a × 1 = 1 × a = a. One changes nothing.

4.4 Subtraction and division are NOT associative

$$(20 - 10) - 5 = 10 - 5 = 5, \qquad 20 - (10 - 5) = 20 - 5 = 15.$$

Two different totals from the same three numbers — the bracket move fails.

$$(16 \div 4) \div 2 = 4 \div 2 = 2, \qquad 16 \div (4 \div 2) = 16 \div 2 = 8.$$

Again, two different answers — the law does not hold.

Moving the brackets in subtraction or division changes the answer. BODMAS exists partly because of this: the position of the brackets is meaningful for these operations.

5. The Distributive Property — Hand the Multiplier Out

5.1 The rule

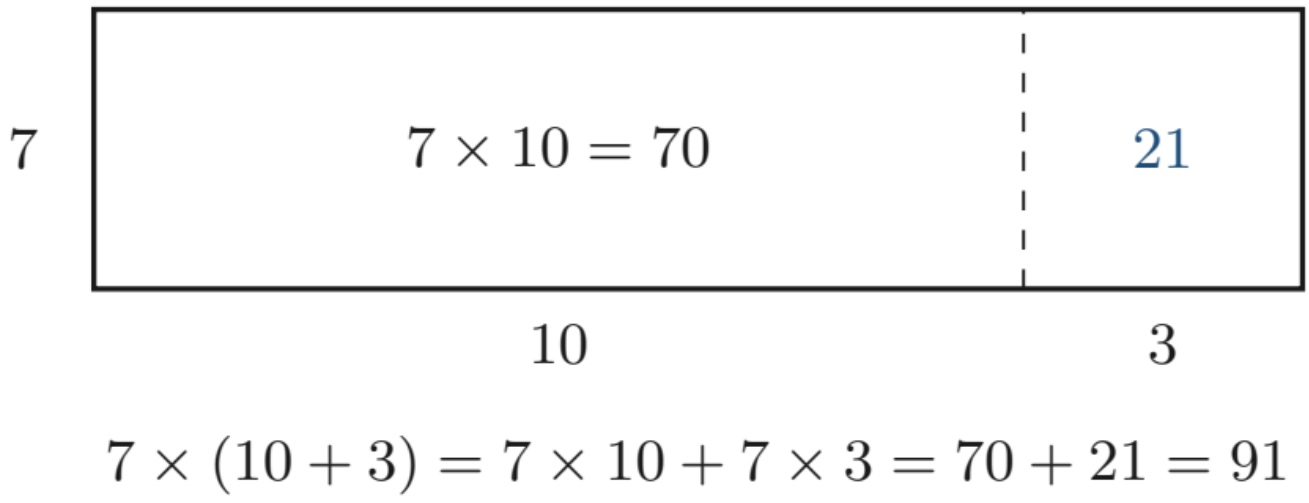
For any numbers a, b, c:

$$a \times (b + c) = a \times b + a \times c$$
$$a \times (b - c) = a \times b - a \times c$$

The multiplier a does not see (b + c) as one lump — it distributes itself to each member of the bracket.

5.2 The rectangle picture

See Figure 1: one big rectangle of base 10 + 3 and height 7 has the same area as the two smaller rectangles 7 × 10 and 7 × 3 glued side by side. The distributive property is really a statement about cutting rectangles.



$$7 \times (10 + 3) = 7 \times 10 + 7 \times 3 = 70 + 21 = 91$$

5.3 Working both directions

- Expanding (bracket side): 7 × 13 = 7 × (10 + 3) = 70 + 21 = 91.
- Collecting (product side — used far more in algebra): 36 + 4 = 4 × 9 + 4 × 1 = 4 × (9 + 1) = 40.

6. Mental Arithmetic with the Laws

This is the section that turns the laws from theory into marks on a speed test.

6.1 Make the addition round (commutative + associative)

For 199 + 364: swap so the awkward 199 becomes 200, add, then correct the extra 1:

$$199 + 364 = (200 + 364) - 1 = 564 - 1 = 563.$$

6.2 Re-group the multiplication (associative)

Multiply in whatever order makes a round product. Learn these friends: 2 × 5 = 10, 4 × 25 = 100, 8 × 125 = 1 000.

$$25 \times 16 = 25 \times (4 \times 4) = (25 \times 4) \times 4 = 100 \times 4 = 400.$$

$$8 \times 43 \times 125 = (8 \times 125) \times 43 = 1\,000 \times 43 = 43\,000.$$

6.3 Split a factor near a round number (distributive)

$$98 \times 7 = (100 - 2) \times 7 = 700 - 14 = 686.$$

$$501 \times 12 = (500 + 1) \times 12 = 6\,000 + 12 = 6\,012.$$

Exam habit: when a question says "without a calculator" or "use suitable properties", the examiner is telling you the intended route — a law, not long column arithmetic.

7. Fifteen Fully Worked Examples (Mixed Angles)

Example 1 — Spot the swapped sum

Ara's working shows 345 + 78 = 78 + 345. Which property makes this true?

Both sides equal 423. Swapping the two addends changed nothing, and it would work for any two numbers.

Answer: The commutative property of addition.

Example 2 — Spot the swapped product

Show that 4 × 927 = 927 × 4 without long multiplication.

4 × 927: split 927 = 900 + 27 (distributive): 3 600 + 108 = 3 708. 927 × 4 = 927 + 927 + 927 + 927 = 3 708 (repeated addition, same total).

Answer: Both are 3 708 — commutative property of multiplication.

Example 3 — Moving brackets in addition

Evaluate (2 + 19) + 81 and 2 + (19 + 81).

First: 21 + 81 = 102. Second: 2 + 100 = 102.

Answer: Both = 102 — the associative property of addition. The second grouping was faster because 19 + 81 makes 100.

Example 4 — Why subtraction cannot commute

Kwame says: "It never matters which way you compute: 9 - 4 or 4 - 9." Correct him with a counterexample.

9 - 4 = 5, but 4 - 9 = −5. Different answers, so subtraction is not commutative. One counterexample is enough to disprove a claimed property.

Example 5 — Why division cannot be regrouped

Evaluate (16 ÷ 4) ÷ 2 and 16 ÷ (4 ÷ 2).

(16 ÷ 4) ÷ 2 = 4 ÷ 2 = 2. But 16 ÷ (4 ÷ 2) = 16 ÷ 2 = 8.

Answer: 2 ≠ 8, so division is not associative. Brackets are compulsory in division.

Example 6 — Distributive into a sum

Work out 7 × (10 + 3) in two ways.

Way 1 (distribute): 7 × 10 + 7 × 3 = 70 + 21 = 91. Way 2 (bracket first): 7 × 13 = 91.

Answer: 91 both ways — same property, same result.

Example 7 — Distributive into a difference

Work out 6 × (50 - 2) by distributing.

6 × 50 - 6 × 2 = 300 - 12 = 288. (Check: 6 × 48 = 288, correct.)

Example 8 — Distributive in three parts

Work out 8 × (100 + 30 + 7).

800 + 240 + 56 = 1 096.

Answer: 1 096 (= 8 × 137). The multiplier may be handed out to any number of parts of the bracket.

Example 9 — The mystery bracket

Find ? if 7 × (12 + ?) = 84 + 35.

Distribute the left side: 7 × 12 + 7 × ? = 84 + 7 ?. Compare with the right side: 84 + 7? = 84 + 35 → 7? = 35 → ? = 5. (Check: 7 × 17 = 119 and 84 + 35 = 119 — correct.)

Example 10 — Round-up-and-correct addition

Mentally compute 199 + 364.

(200 + 364) - 1 = 564 - 1 = 563.

Answer: 563. Used laws: commutative + associative to reach the friendly 200.

Example 11 — Friendly pair re-grouping

Mentally compute 25 × 16.

16 = 4 × 4, so 25 × (4 × 4) = (25 × 4) × 4 = 100 × 4 = 400.

Answer: 400 — associative property chosen so a round 100 appears first.

Example 12 — Three-factor re-grouping

Mentally compute 8 × 43 × 125.

Re-order and re-group: (8 × 125) × 43 = 1 000 × 43 = 43 000.

Answer: 43 000 (uses commutative + associative of multiplication).

Example 13 — Near-round multiplier

Mentally compute 98 × 7.

(100 - 2) × 7 = 700 - 14 = 686.

Answer: 686 — distributive property over subtraction.

Example 14 — Authentic BECE 1998: name the property

State the property used in the statement p(q + r) = pq + pr.

The multiplier p on the outside of the bracket has been handed out to both q and r inside it, becoming pq + pr.

Answer: The distributive property (official key: distributive). (Bank item `Q_(d)b644641d061`.)

Example 15 — Bridge to algebra (mock-linked)

Expand and simplify, using the distributive property:

(a) 5(3 - y) — the shape of Mock P1 Q18's first move. 5 × 3 - 5 × y = 15 - 5y.

(b) 2(3x + 3) — the shape of Mock P2 Q4(d)'s inner step. 2 × 3x + 2 × 3 = 6x + 6.

(c) Stretch (authentic BECE 2009, `Q_(c)bcf875f87f8`): solve 2x - 3(x - 1) = 6. Distribute: 2x - 3x + 3 = 6 → −x + 3 = 6 → x = −3 (official key: −3). Full equation-solving technique is delivered in the Equations & Inequalities lesson — here, notice that the only new move was the distributive property.

8. Exercise Sets (5 × 5, No MCQ)

Instructions: write full workings for every question. Where a question asks you to state a property, name it and say briefly why. Calculator use is not needed anywhere.

Exercise 1 — Name that property

1. 7 + 519 = 519 + 7. What property does this show?
2. (8 × 5) × 20 = 8 × (5 × 20). What property does this show?
3. 12 × (30 + 4) = 360 + 48. What property turns the left side into the right side?
4. Explain, with numbers, why 9 - 4 is not equal to 4 - 9. Which property is missing in subtraction?
5. Find the missing number so that the statement stays true for any value: ? × 35 = 35 × 72. Then name the property that guarantees your answer is the only possible one.

Exercise 2 — True or false (justify each)

1. "Multiplication is commutative." True or false? Give a reason or example.
2. "Subtraction is associative." True or false? Justify with (20 - 10) - 5 vs 20 - (10 - 5).
3. a × (b + c) = a × b + a × c for all numbers a, b, c. True or false?
4. 12 ÷ 3 = 3 ÷ 12. True or false? Compute both sides to justify.
5. Adding zero changes nothing; multiplying by one changes nothing. State the names of these two properties.

Exercise 3 — Mental arithmetic with the laws

Compute mentally, and write one sentence naming the properties you used.

1. 199 + 450
2. 25 × 36
3. 8 × 34 × 125
4. 97 × 4
5. A trader buys 98 exercise books at GH₵ 4 each. Use the distributive property to find the total cost mentally.

Exercise 4 — Fill the gaps

1. Complete: 6 × (12 + ?) = 72 + 30.
2. Complete: 9 × (100 - 2) = 900 - ?.
3. Complete: (7 + 8) + 25 = 7 + (? + 25).
4. Using friendly re-grouping, find ? in 4 × 5 × ? = 400.
5. Spiral review (standard form, Day 1): Without long multiplication, use a friendly pair to compute 5 × 2 000 × 8, then write the answer in standard form.

Exercise 5 — Applied problems and the bridge to algebra

1. Expand 5(3 - y) using the distributive property.
2. Expand and simplify 2(3x + 3).
3. [Mock 2026 P1 Q40] A baker has 12 identical cakes. In how many different ways can he divide them up into equal packets? List every way.
4. Spiral review (indices/number operations, Day 3): Evaluate 5 × 2^(3) × 2^(2) in a quick way using re-grouping, then express the answer as a power of 2 combined with 5, and in standard form.
5. Spiral review (standard form, Day 1): Use a friendly pair to compute 25 × 400, and express the answer in standard form.

9. Full Worked Solutions

Exercise 1

1. 7 + 519 = 526 and 519 + 7 = 526. The two addends swapped places without changing the sum. Property: commutative property of addition.

2. Both sides regroup the three factors. Left: 40 × 20 = 800. Right: 8 × 100 = 800. Property: associative property of multiplication.

3. The multiplier 12 has been handed out to 30 and to 4: 360 + 48. Property: distributive property.

4. 9 - 4 = 5 but 4 - 9 = −5. Different answers, so the operations cannot be swapped freely. Subtraction is missing the commutative property.

5. By the commutative property of multiplication, a × b = b × a is true for every pair of numbers, so the other factor must be 72: the blank is 72. The guarantee is the commutative property — whatever number multiplies 35 on the left must be the number 35 multiplies on the right.

Exercise 2

1. True. a × b = b × a always; e.g. 4 × 11 = 11 × 4 = 44.

2. False. (20 - 10) - 5 = 5, but 20 - (10 - 5) = 15. The two groupings give different results, so moving brackets is not allowed in subtraction.

3. True. This is the distributive property — true for all numbers (the rectangle picture guarantees it).

4. False. 12 ÷ 3 = 4 but 3 ÷ 12 = 0.25. Division is not commutative — swapping divisor and dividend changes the answer.

5. Zero is the additive identity (a + 0 = a); one is the multiplicative identity (a × 1 = a).

Exercise 3

1. 199 + 450 = (200 + 450) - 1 = 650 - 1 = 649. Used: commutative + associative to reach the round 200, then corrected by the extra 1.

2. 25 × 36 = 25 × (4 × 9) = (25 × 4) × 9 = 100 × 9 = 900. Used: factorising 36 = 4 × 9, then the associative property for the friendly 100.

3. 8 × 34 × 125 = (8 × 125) × 34 = 1 000 × 34 = 34 000. Used: commutative property to place 8 × 125 together, then associativity.

4. 97 × 4 = (100 - 3) × 4 = 400 - 12 = 388. Used: writing 97 = 100 - 3, then the distributive property over subtraction.

5. Total = 98 × 4, and 98 = 100 - 2: (100 - 2) × 4 = 400 - 8 = GH₵ 392. Used: distributive property over subtraction.

Exercise 4

1. 6 × (12 + ?) = 6 × 12 + 6? = 72 + 6?. Compare with 72 + 30: 6? = 30 → ? = 5. (Check: 6 × 17 = 102 and 72 + 30 = 102 — correct.)

2. 9 × (100 - 2) = 900 - 9 × 2 = 900 - 18.

3. The associative property keeps the values in place and only moves the brackets, so the blank is 8.

4. Friendly re-grouping: we want two of the three factors to give a round product. 4 × 5 = 20 is not round, so choose ? = 20 so that 5 × 20 = 100 can be grouped next to the remaining factor: 4 × (5 × 20) = 4 × 100 = 400, check confirmed, so ? = 20.

5. Friendly pair: 5 × 8 = 40, so 5 × 2 000 × 8 = (5 × 8) × 2 000 = 40 × 2 000 = 80 000. Standard form: 80 000 = 8 × 10^(4).

Exercise 5

1. 5(3 - y) = 5 × 3 - 5 × y = 15 - 5y (distributive property).

2. 2(3x + 3) = 2 × 3x + 2 × 3 = 6x + 6.

3. Equal packets of size k means k is a factor of 12. Bonus method (distributive thinking): if 12 cakes split into p packets with k cakes each, then p × k = 12. Factor pairs of 12: 1 × 12, 2 × 6, 3 × 4, 4 × 3, 6 × 2, 12 × 1. Distinct equal-packet groupings: 12 packets of 1; 6 of 2; 4 of 3; 3 of 4; 2 of 6; 1 packet of 12.

Answer: 6 ways (matches the official key C).

4. Multiply the powers of 2 first (associative cs.): 5 × (2^(3) × 2^(2)) = 5 × 2^(5) = 5 × 32 = 160. As an expression: 5 × 2^(5); in standard form: 160 = 1.6 × 10^(2).

5. Friendly pair: 25 × 4 = 100, and 400 = 4 × 100, so 25 × 400 = 25 × (4 × 100) = (25 × 4) × 100 = 10 000. Standard form: 10 000 = 1 × 10^(4).

10. Bank and Mock Links

Item	ID	Source	Where used
State the property in p(q+r) = pq+pr (BECE 1998)	`Q_(d)b644641d061`	BECE 1990–2013 compilation	Example 14
Solve 2x - 3(x-1) = 6 (BECE 2009)	`Q_(c)bcf875f87f8`	BECE 1990–2013 compilation	Example 15(c)
12 cakes into equal packets (Mock 2026 P1 Q40)	`Q_(d)5a5e61b82f1`	user mock 0201-026/27	Exercise 5, Q3

11. Lesson Summary (Cheat Sheet)

Law	Statement	Works for	Fails for
Commutative	a + b = b + a; a × b = b × a	+ and ×	- and ÷
Associative	(a+b)+c = a+(b+c); (ab)c = a(bc)	+ and ×	- and ÷
Distributive	a(b ± c) = ab ± ac	bridges × into +/-	—
Identities	a + 0 = a; a × 1 = a	all	—

Mental maths trio: make the addition round (swap + re-group), make the product round (friendly pairs: 4×25, 8×125), split the near-round factor (distribute 100 - 2 style).

Lesson prepared under `Grade9-Maths/Rules.md` — plain English, British notation, full workings for every question. Quota override in force for this round: 5 sections × 5 questions.